

NOTES FOR MATH 3150, WEEK 1

1. NOTATION

1.1. **Operations with sets.** Throughout these notes:

A, B, C, S, T denote sets of numbers.

x, y, z, a, b denote elements of these sets.

$x \in A$ means that x is in A .

$A \subset B$ means that A is included in B , so if $x \in A$ then $x \in B$ as well.

$A \cap B$ is the set of all x both in A and in B , so

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

$A \cup B$ is the set of all x either in A or in B , so

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

$A \setminus B$ is the set of all x that are in A but not in B , so

$$A \setminus B = \{x : x \in A \text{ but } x \notin B\}.$$

The set $A^c = \mathbb{R} \setminus A$ is called the complement of A .

1.2. **Functions.** Functions are denoted with $f, g : D \rightarrow \mathbb{R}$. The set D is called the domain of f . The set

$$f(D) = \{y \in \mathbb{R} : y = f(x), \text{ for some } x \in D\}$$

is called the range of f .

A function is called one-to-one if

$$f(x) = f(y) \text{ implies } x = y.$$

A function $f : D \rightarrow G$ is called onto if the range of f is equal to G ; that is,

$$f(D) = G$$

2. THE NATURAL NUMBERS $\mathbb{N}$

We use axioms to abstractly define natural numbers.

We denote with $\mathbb{N} = \{1, 2, 3, \dots\}$.

2.1. The Peano axioms. These are the axioms of natural numbers.

N1. $1 \in \mathbb{N}$

N2. If $n \in \mathbb{N}$, then $n + 1 \in \mathbb{N}$. We say $n + 1$ is the successor of n .

N3. 1 is not the successor of any element in $\mathbb{N}$.

N4. If $n, m \in \mathbb{N}$ have the same successor, then $n = m$.

N5. If $S \subset \mathbb{N}$ is a subset with the properties

- $1 \in S$
- If $n \in S$, then $n + 1 \in S$

then $S = \mathbb{N}$.

These axioms imply all the known properties of natural numbers. The last one, N5, is the basis of mathematical induction, a very important mathematical principle.

Mathematical induction:

Let $P_1, P_2, \dots, P_n, \dots$ be a list of statements (propositions). Assume that

- P_1 is true (called the basis of induction).
- If P_n is true, then P_{n+1} is true as well (called the induction step).

Then P_n is true for all $n \in \mathbb{N}$.

Examples:

1. *An identity.* Prove that for all $q \neq 1$,

$$\sum_{k=0}^n q^k = \frac{1 - q^{n+1}}{1 - q}.$$

Proof. First, let us recall that we are using the summation notation:

$$\sum_{k=0}^n a_k = a_0 + a_1 + \dots + a_n.$$

Let P_n be the statement

$$\sum_{k=0}^n q^k = \frac{1 - q^{n+1}}{1 - q}.$$

We prove P_n is true for all $n \in \mathbb{N}$.

Basis of induction: $n = 1$, then P_1 is simply saying that

$$1 + q = \frac{1 - q^2}{1 - q},$$

which is clearly true.

Induction step: Assume that P_n is true, prove that P_{n+1} is true. So we know that

$$\sum_{k=0}^n q^k = \frac{1 - q^{n+1}}{1 - q}.$$

and wish to prove that

$$\sum_{k=0}^{n+1} q^k = \frac{1 - q^{n+2}}{1 - q}.$$

But

$$\begin{aligned} \sum_{k=0}^{n+1} q^k &= \sum_{k=0}^n q^k + q^{n+1} \\ &= \frac{1 - q^{n+1}}{1 - q} + q^{n+1} \\ &= \frac{1 - q^{n+1} + q^{n+1}(1 - q)}{1 - q} \\ &= \frac{1 - q^{n+2}}{1 - q}, \end{aligned}$$

where in the second line we used that P_n is true.

This finishes the induction proof that

$$\sum_{k=0}^n q^k = \frac{1 - q^{n+1}}{1 - q}.$$

□

2. *An inequality.* Prove that

$$\sum_{k=1}^n \frac{1}{k^3} < \frac{3}{2} - \frac{1}{n},$$

for all $n \geq 3$.

Proof. Let P_n be the statement

$$\sum_{k=1}^n \frac{1}{k^3} < \frac{3}{2} - \frac{1}{n}.$$

We use induction to prove that P_n is true for all $n \geq 3$.

Basis of induction: P_3 means that $1 + \frac{1}{2^3} + \frac{1}{3^3} < \frac{3}{2} - \frac{1}{3}$, which is the same as $\frac{1}{27} < \frac{1}{24}$.

Induction step. Assume that P_n is true and prove P_{n+1} is true as well.
We know that

$$\sum_{k=1}^n \frac{1}{k^3} < \frac{3}{2} - \frac{1}{n}.$$

and wish to prove that

$$\sum_{k=1}^{n+1} \frac{1}{k^3} < \frac{3}{2} - \frac{1}{n+1}.$$

However,

$$\begin{aligned} \sum_{k=1}^{n+1} \frac{1}{k^3} &= \sum_{k=1}^n \frac{1}{k^3} + \frac{1}{(n+1)^3} \\ &< \frac{3}{2} - \frac{1}{n} + \frac{1}{(n+1)^3}, \end{aligned}$$

where in the second line we used that P_n is true.

Now to prove P_{n+1} , we simply check that

$$\frac{3}{2} - \frac{1}{n} + \frac{1}{(n+1)^3} < \frac{3}{2} - \frac{1}{n+1}.$$

This is equivalent to

$$\frac{1}{(n+1)^3} < \frac{1}{n(n+1)},$$

which after simplification is

$$n < (n+1)^2.$$

Since $n^2 + n + 1 > 0$ for all $n \geq 3$, this proves P_{n+1} . This ends the induction proof that

$$\sum_{k=1}^n \frac{1}{k^3} < \frac{3}{2} - \frac{1}{n},$$

for all $n \geq 3$. □

Remarks:

- (a). The induction can start from $n \geq n_0$.
- (b). We would not be able prove by induction that

$$\sum_{k=1}^n \frac{1}{k^3} < \frac{3}{2},$$

although this inequality is weaker than the one we proved by induction. Check for yourselves what breaks down in proving the above inequality by induction.

3. *The binomial theorem.* Prove that for any $n \in \mathbb{N}$ and any $a, b \in \mathbb{R}$ we have

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k},$$

where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

and for $m \in \mathbb{N}$ we denote the factorial by $m! = 1 \cdot 2 \cdot \dots \cdot m$.

This is left to the reader to prove. In the induction step, we end up using the following identity:

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}.$$

You can prove this identity directly from the definition of $\binom{n}{k}$.

4. *A non-example.*

Let us attempt to prove by induction that

$$n^2 + 5n + 1$$

is an even number for all $n \geq 1$.

As we have seen, the step P_n true implies P_{n+1} true is the "serious" one, so let's check that first.

Assume P_n is true, that is $n^2 + 5n + 1$ is an even number. Let us prove P_{n+1} is true, i.e. $(n+1)^2 + 5(n+1) + 1$ is an even number.

But

$$(n+1)^2 + 5(n+1) + 1 = (n^2 + 5n + 1) + 2(n+3),$$

so it's obviously an even number, being the sum of two even numbers.

However, P_n is never true! Indeed, if n is even, then $n^2 + 5n$ is even, so $n^2 + 5n + 1$ is odd.

Similarly for n odd, or simply use the above.

Importantly, in this example, the base step does not hold!

3. THE RATIONAL NUMBERS $\mathbb{Q}$

By definition, $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is the set of integers. So, $\mathbb{Z}$ contains all natural numbers, their negatives and zero.

By definition, $\mathbb{Q} = \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}$.

Let us note some basic algebraic properties of $\mathbb{Q}$.

For all $a, b, c \in \mathbb{Q}$ we have the following:

Additive properties:

A1. $a + (b + c) = (a + b) + c$

A2. $a + b = b + a$

A3. $a + 0 = a$

A4. for each $a \in \mathbb{Q}$ there exists $-a \in \mathbb{Q}$ so that $a + (-a) = 0$

Multiplicative properties:

M1. $a(bc) = (ab)c$

M2. $ab = ba$

M3. $a1 = a$

M4. for each $a \neq 0$ there exists $\frac{1}{a}$ so that $a\frac{1}{a} = 1$.

Distributive properties:

DL. $a(b + c) = ab + ac$.

Besides $+$ and $\cdot$, $\mathbb{Q}$ has an additional structure, it has an order. Given $a, b, c \in \mathbb{Q}$ we have

Order properties:

O1. Either $a \leq b$ or $b \leq a$.

O2. If $a \leq b$ and $b \leq a$ then $a = b$

O3. If $a \leq b$ and $b \leq c$ then $a \leq c$

O4. If $a \leq b$ then $a + c \leq b + c$

O5. If $a \leq b$ and $0 \leq c$ then $ac \leq bc$.

3.1. **(Not to be tested).** The basic axioms of numbers described in A, M, DL and O imply many others that we are familiar with. For example, we have the following.

Proposition 1. For any $a, b \in \mathbb{Q}$ we have

(a) $a \cdot 0 = 0$

(b) $(-a)(-b) = ab$

(c) $0 \leq a^2$

(d) $0 \leq 1$

Proof. (a) We have that $0 + 0 = 0$ by A3. Now multiply by a and use DL. It follows that

$$a \cdot 0 + a \cdot 0 = a \cdot 0.$$

So the rational number $c = a \cdot 0$ satisfies

$$c + c = c.$$

By A4 there exists $-c$ so that

$$c + (-c) = 0.$$

Now from

$$c + c = c \quad \text{and} \quad c + (-c) = 0$$

we get

$$\begin{aligned} c &= c + 0 \\ &= c + c + (-c) \\ &= c + (-c) \\ &= 0. \end{aligned}$$

(b) Note that

$$(-1)a = -a,$$

because

$$\begin{aligned} a + (-1)a &= 1 \cdot a + (-1) \cdot a \\ &= (1 + (-1))a \\ &= 0 \cdot a \\ &= 0. \end{aligned}$$

Let us prove first that

$$(-1)(-1) = 1.$$

Indeed,

$$\begin{aligned} 0 &= (-1) \cdot 0 \\ &= (-1) \cdot ((-1) + 1) \\ &= (-1) \cdot (-1) + (-1) \cdot 1 \\ &= (-1) \cdot (-1) + (-1). \end{aligned}$$

So $(-1) \cdot (-1)$ is the inverse of (-1) , therefore

$$(-1) \cdot (-1) = 1.$$

Finally, we have

$$\begin{aligned} (-a)(-b) &= (-1)(-1)ab \\ &= 1ab \\ &= ab, \end{aligned}$$

which is what we wanted.

(c) If $0 \leq a$ then use O5 to get $0 \leq a \cdot a$.

If $a \leq 0$, then let

$$b = -a.$$

We note that

$$0 \leq b.$$

Indeed, by O4 we have

$$a + (-a) \leq 0 + (-a),$$

therefore $0 \leq -a$.

Now we get

$$\begin{aligned} a^2 &= (-b)^2 \\ &= b^2 \\ &\geq 0 \end{aligned}$$

where in the second line we used (b).

(d). Since $1 = 1^2$, it follows that $0 \leq 1$.

□